AIORI HACKATHON 3 Category: Cybersecurity • Problem Statement: PS-019 Team Code: A3-PS019-TC155

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Personal-Data Discovery and Privacy Evidence Inventory

A configurable scanner inspecting synthetic structured datasets (CSV, SQLite) to identify candidate personal-data fields, link them to declared business purposes and retention metadata, and output an audited technical evidence inventory for human review.

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Problem Description

Stored personal data is hard to find, and its purpose and retention are rarely documented.

Dark Data Accumulation

• Personal data (PII) builds up in unindexed database tables and CSV spreadsheets over time.

• Manual data discovery audits are painfully slow and regularly miss nested or renamed fields.

Missing Metadata Gap

• Data protection laws mandate linking each stored item to a declared purpose & retention rule.

• In practice, discovery outputs are rarely connected to enterprise retention schedules.

The Privexis Solution

• Inspects synthetic CSVs & SQLite across 5 predefined personal-data classes.

• Automates linking candidate fields to declared business purpose & retention metadata.

Clear Scope Boundary: Outputs technical evidence only. Makes no automatic legal decisions or compliance determinations.

Synthetic Data Only

Background

Finding personal data is the first step to governing it.

STATUTORY GROUNDING: DPDP ACT, 2023 [1]

India's Digital Personal Data Protection Act, 2023 ties personal data to a specified purpose and expects erasure once that purpose is no longer served.

Specified Purpose
Personal data may only be processed for transparent, declared objectives.

Storage Limitation
Data must not be retained once the processing rationale expires.

The Governance Requirement

A retention schedule records how long each category of data is kept, who owns it, and what happens at the end of its life. Because an organization cannot link purpose and retention to a field it has not discovered, automated discovery must precede governance.

SCOPE & RESPONSIBILITY BOUNDARIES

System Scope (Privexis)
Produces reproducible technical evidence: detected candidate data classes, syntax matches, validation checks, and metadata gap flags.

Human Reviewer Scope
Reviews evidence inventory, validates false positives, decides lawful-basis determinations, and authorizes remediation workflows.

Literature & RFC Review

Standards and tools our design builds on.

Reference Standard	Technical Domain	Core Principle / Capability	How Privexis Implements It
DPDP Act, 2023 [1]	**Statutory Framework**	Mandates purpose specification and storage limitation upon objective completion.	Defines purpose and retention metadata fields in inventory.
Verhoeff Checksum [2]	**Dihedral Group D5**	Detects all single-digit errors and all adjacent transposition errors.	Validates 12-digit Aadhaar-style synthetic numbers.
RFC 5322 [3]	**Internet Syntax**	Defines standard syntax grammar for Internet email message addr-spec.	Syntactic regex baseline for the email detection rule.
Microsoft Presidio [4]	**Discovery Framework**	Pattern recognizers and NER context enhancers for free-form text.	Pattern + NER detection idea; we add metadata linking & gap reports.
spaCy NER [5]	**Statistical NLP**	Industrial-strength named entity recognition models.	Optional PERSON detection for candidate name columns.
Netwrix PII Guide [6]	**Security Taxonomy**	Taxonomy of rule-based vs machine learning data discovery techniques.	Overview of rule-based and ML detection trade-offs.

📖 Privexis bridges syntactic detection standards (RFC 5322, Verhoeff) with operational privacy governance (DPDP Act, 2023).

Peer & Statutory Standards

Proposed Solution

Privexis: a configurable scanner that turns raw files into a reviewable evidence inventory.

Dual Source Connectors & Bounded Sampler

Connectors read CSV files and one SQLite database; a sampler extracts a bounded sample of values per column to ensure rapid, deterministic scanning.

Declarative YAML Rule Engine

A YAML rule engine scores each column by fusing column-name hints, regex value patterns, and mathematical checksum validators.

Metadata Linker & Automated Gap Report

A linker joins each flagged field to its declared purpose and retention entry. A gap report systematically flags fields with missing metadata.

Candidate Evidence for Human Review

Every discovery output is explicitly tagged as candidate evidence for human review, preventing automated misclassification.

5 Data Classes (PS-019)

Mandatory

• **Name**	Alphabetic / NER
• **Email address**	RFC 5322 regex
• **Phone number**	10-digit / E.164
• **Government ID**	**Aadhaar (Verhoeff) / PAN**
• **Date of birth (DOB)**	ISO & DMY bounds



Optimization Proposed

Before and after: what each design choice improves.

Area	Before	After (proposed)
Detection	Column-name match only	**Name hints + value patterns + validators, combined into one confidence score**
Rules	Hard-coded in the script	**Editable YAML, no code change needed**
Metadata	Purpose and retention scattered or undocumented	**Linked to each field in one inventory**
Gaps	Found by manual audit	**Missing purpose or retention flagged automatically**
Quality	Not measured	**Per-class precision and recall on a hand-labelled set**

Key Engineering Impact: Eliminates hard-coded assumptions by decoupling detection logic into declarative configuration and formal metric evaluation.

Optimization Matrix

Solution Architecture & Design

Component view: files and rules go in, evidence reports come out.

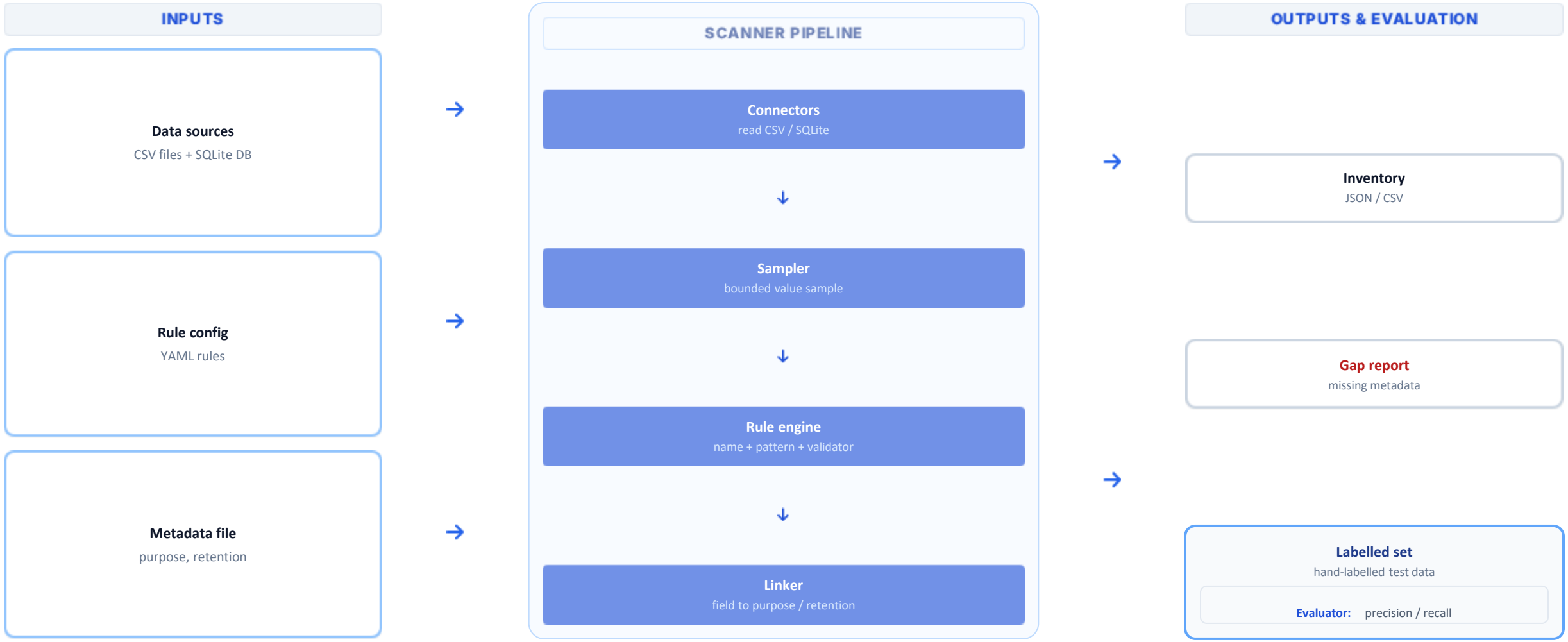


Figure: System Architecture UML Diagram

Data Flow Diagram

Level-1 view: who supplies data, how it is processed, and what is stored.

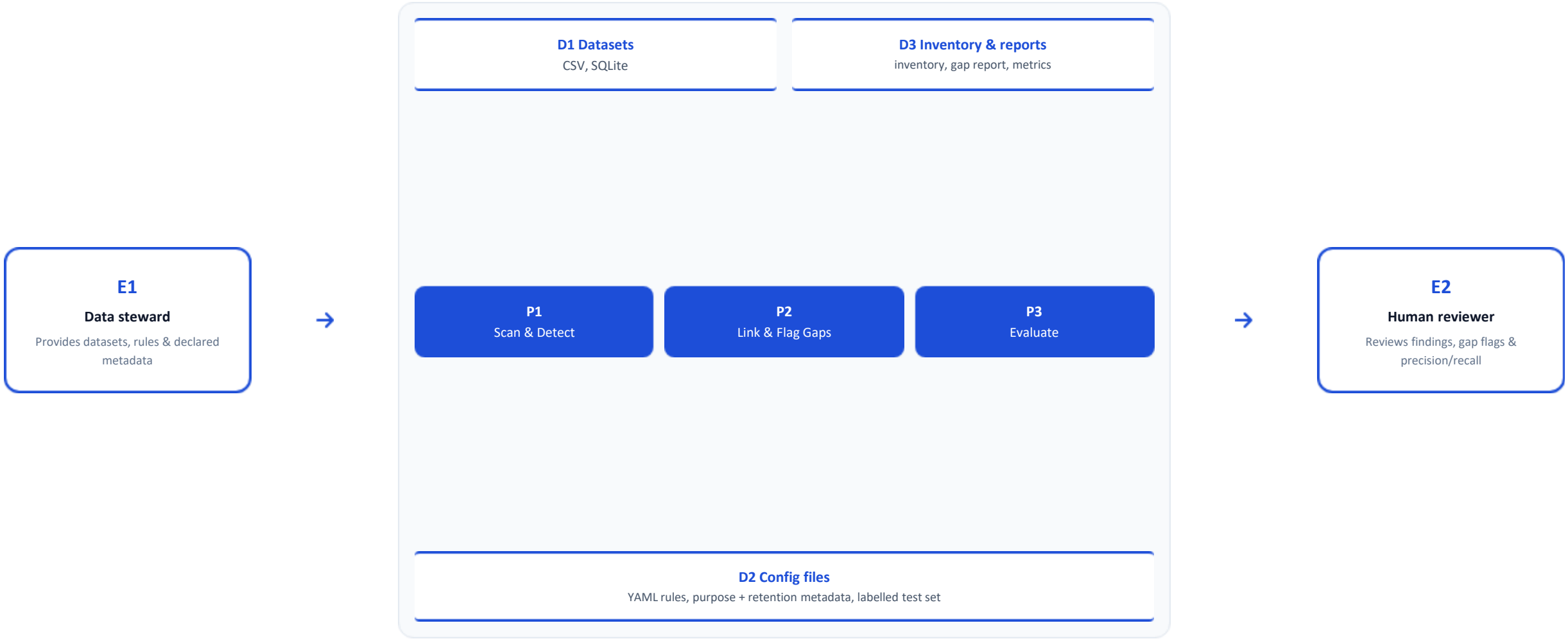


Figure: Data Flow Diagram (DFD)

Detection Pipeline & Evidence Inventory

Multimodal scoring and illustrative technical evidence inventory.

DETECTION PIPELINE: 5-STEP MULTIMODAL VERIFICATION

1. Column Hint	2. Value Pattern	3. Strict Validator	4. Score Engine	5. Inventory Output
Token substring match	Regex pattern check	Verhoeff / RFC 5322	Confidence scoring	Logged for human triage

ILLUSTRATIVE TECHNICAL EVIDENCE INVENTORY (SYNTHETIC DATA EXAMPLE)

**Illustrative schema; not experimental validation data*

Source / Table	Column Name	Candidate Class	Detection Evidence	Declared Purpose	Retention Schedule	Review Status
users.sqlite / acc	usr_contact	Phone number	10-digit regex match	2FA Security	Closure + 180d	V Verified Candidate
onboarding.csv	national_id	Govt ID (Aadhaar)	Verhoeff checksum passed	KYC Identity	Statutory (7 yrs)	V Verified Candidate
temp_dump.csv	tmp_mail	Email address	RFC 5322 regex match	[MISSING PURPOSE]	[MISSING RETENTION]	E EVIDENCE GAP FLAG

🚩 **Missing Metadata Flag:** Automatically surfaces unlinked fields in the gap report to assist human privacy stewards.

Structured Inventory Format

Deployment Timeline

Five milestones from setup to demo.

Setup

Repo, access, synthetic CSV + SQLite data

Scanner

Connectors, sampler, YAML rule engine

Linking

Purpose and retention mapping, gap report

Evaluation

Labelled set, per-class precision and recall

5

Delivery

PDF, team video, demo, code submission

Milestone 05

Methodology:

Bibliography & Operational Boundaries

Authoritative references and PS-019 governance compliance.

BIBLIOGRAPHY REFERENCES

[1] Government of India, MeitY. “The Digital Personal Data Protection Act, 2023 (Act No. 22 of 2023).”

[2] J. Verhoeff. “Error Detecting Decimal Codes.” Mathematical Centre Tract 29, Mathematisch Centrum, Amsterdam, 1969.

[3] P. Resnick (Ed.). “Internet Message Format.” RFC 5322, IETF, 2008.

[4] Microsoft. “Presidio: Data Protection and De-identification SDK.” github.com/microsoft/presidio

[5] Explosion. “spaCy: Named Entity Recognition.” spacy.io

[6] Netwrix. “PII Detection: Why It’s Crucial in Today’s Data Landscape.” Web guide.

PS-019 Traceability Checklist

CSV & SQLite DB:
Dual connectors supported

5 Data Classes:
Configured in YAML rules

Metadata Linker:
Purpose & retention mapping

Quality Evaluation:
Per-class precision & recall

Final Closing Principle

Privexis acts as a technical discovery instrument to produce empirical evidence inventories for human review. It makes no autonomous legal or compliance conclusions.

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